

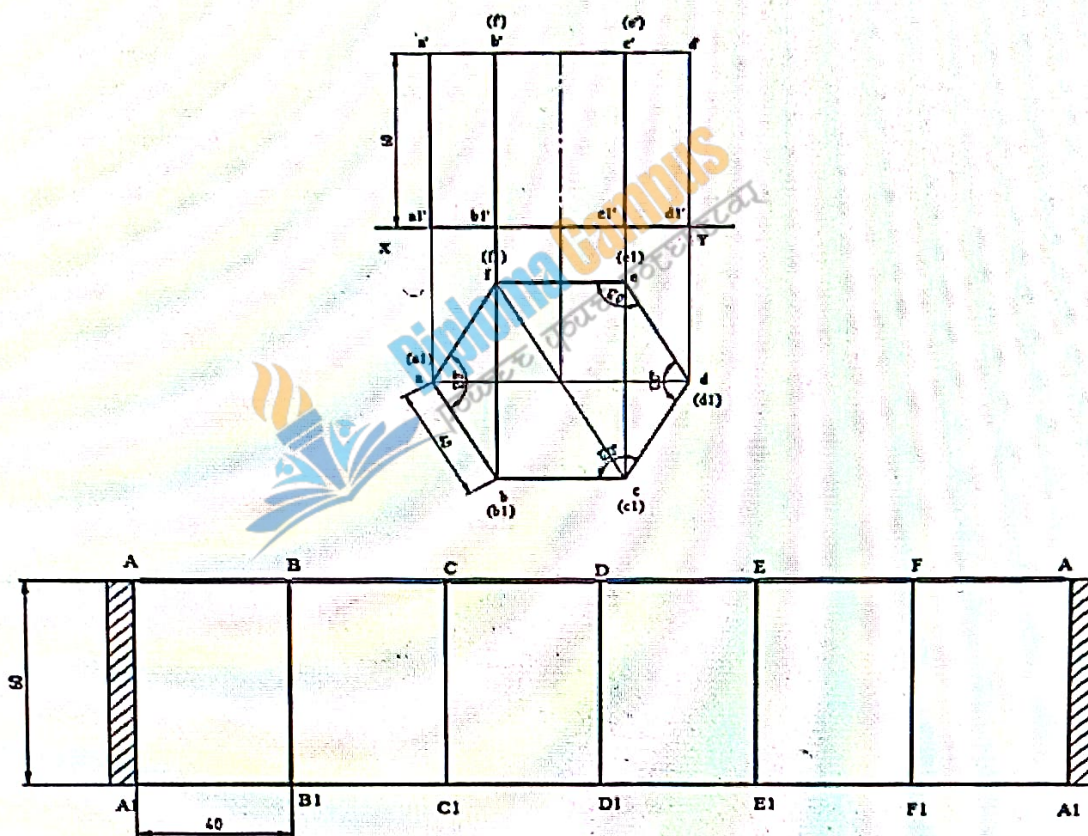
Week-9

1. Explain the Development of Cone and Cylinder
2. Explain the Development of Prism and Pyramid
3. Explain the Development of Funnel

Experiment No.: 01

Date:

Development of Hexagonal Prism



Sequence of operations or procedure for preparing Hexagonal Prism

- Draw the top and front view of the prism of the given dimensions.
- Equal to six sides of the prism $40 \times 6 = 240$ mm.
- Draw a stretch out line A1A1 and mark off the sides of the base along this line i.e. A1B1, B1C1, C1D1 & D1E1, E1F1 and F1A1.
- Erect perpendiculars through these points and mark the edges AA1, BB1, CC1, DD1, EE1, FF1 and AA1 and height of the prism 60mm.
- Add 5mm extra adjacent to AA1 for seam joint.

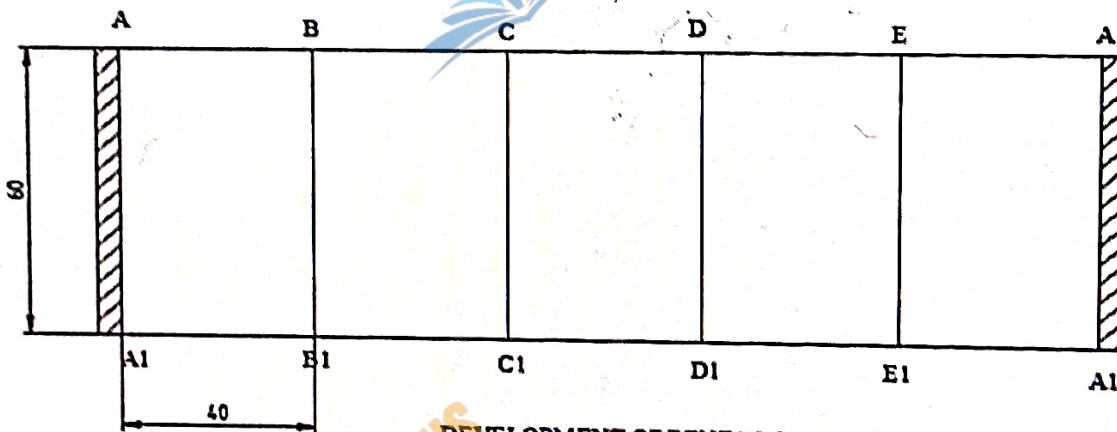
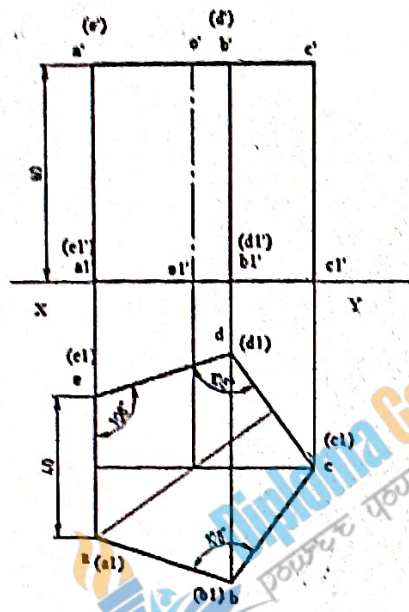


Manufacturing Process

- Cut the development to shape on lines by using paper cutting scissor.
- Place and fix the development on gives G.P sheet by using sticker.
- Punch two points on each bending line.
- Mark the boundary line by using marking pen.
- Remove the paper development join the two points by scriber or marking pen.
- Shear the boundary lines by hand snip.
- Flatten the sheet by using mallet on flatter.
- Always bend the seaming line first and then the remaining lines to get desired shape.
- Finish the surface keeping on stake to shape.
- Solder the joint.
- Finish the surface finishing and complete the job.



Development of Pentagon Prism



DEVELOPMENT OF PENTAGONAL PRISM

All dimensions are in mm

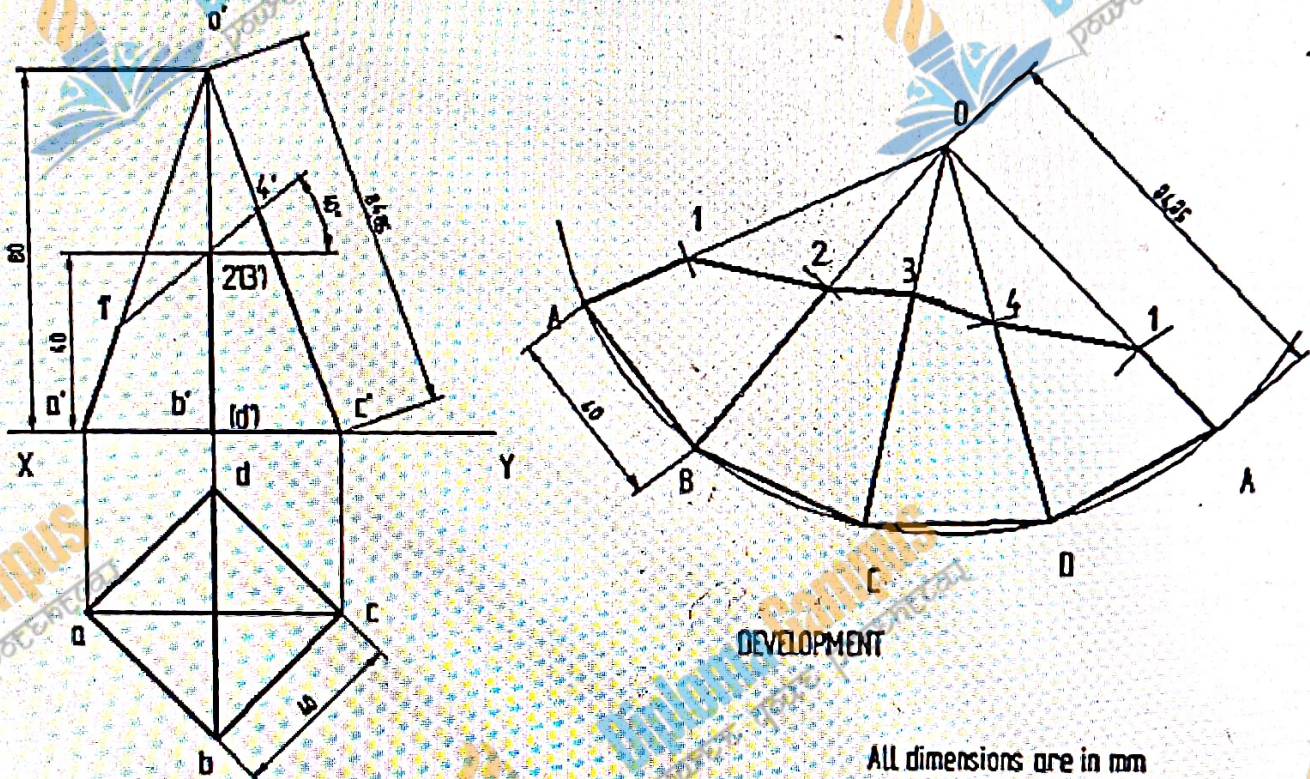


Visual Watermark

Sequence of operations or procedure for preparing Pentagonal Prism

- Draw the Top & Front view of the prism to the given dimensions.
- Draw a stretch out line A_1A_1 and mark off the sides of the base along this line i.e. A_1B_1 , B_1C_1 , C_1D_1 & D_1E_1 and E_1A_1 .
- Erect perpendiculars through these points and mark the edges AA_1 , $B B_1$, CC_1 , DD_1 , EE_1 and AA_1 and height of the prism is 60mm.
- Add 5mm extra adjacent to AA_1 on each side.
- Cut the development to the shape of the line by using paper cutting scissor
- Place & fix the development to the given GP sheet by using sticker.
- Punch the two points on each bending line
- Mark the boundary line by using marking pen.
- Remove the paper development; join the two points by scribe or marking pen.
- Shear the boundary line by hand snip.
- Flatten the sheet by using mallet on flatter.
- Always bend the seaming line first and then the remaining lines to get desired shape.
- Finish the surface keeping the stake to correct shape
- Solder the joints
- Finish the surface finishing and complete the job

Development of Truncated Square Pyramid

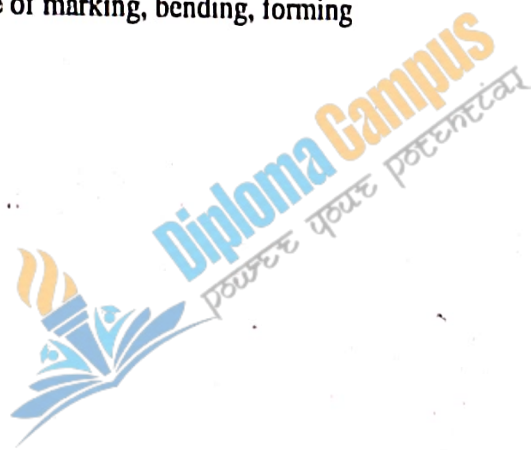


Sequence of operation to prepare truncated square pyramid.

- Draw the front view of the Truncated Pyramid as per dimensions.
- Extend the outer slant edges to mark apex 'O'.
- Project and get the top view of the pyramid and O'C is the true length of the slant edge.
- With O' as center and radius O'C' describe an arc.
- Transfer the true length of top face edges 40mm each on the arc AB, BC, CD and DA. Join them with straight line and to Apex 'O'.
- Section points are marked on the development by measuring their true distances from O on the true length of slant edge in front view.
- Join the sections and draw dark lines to set the development of the surface of the remaining portion of the pyramid.
- Set 5mm extra for seam joint.

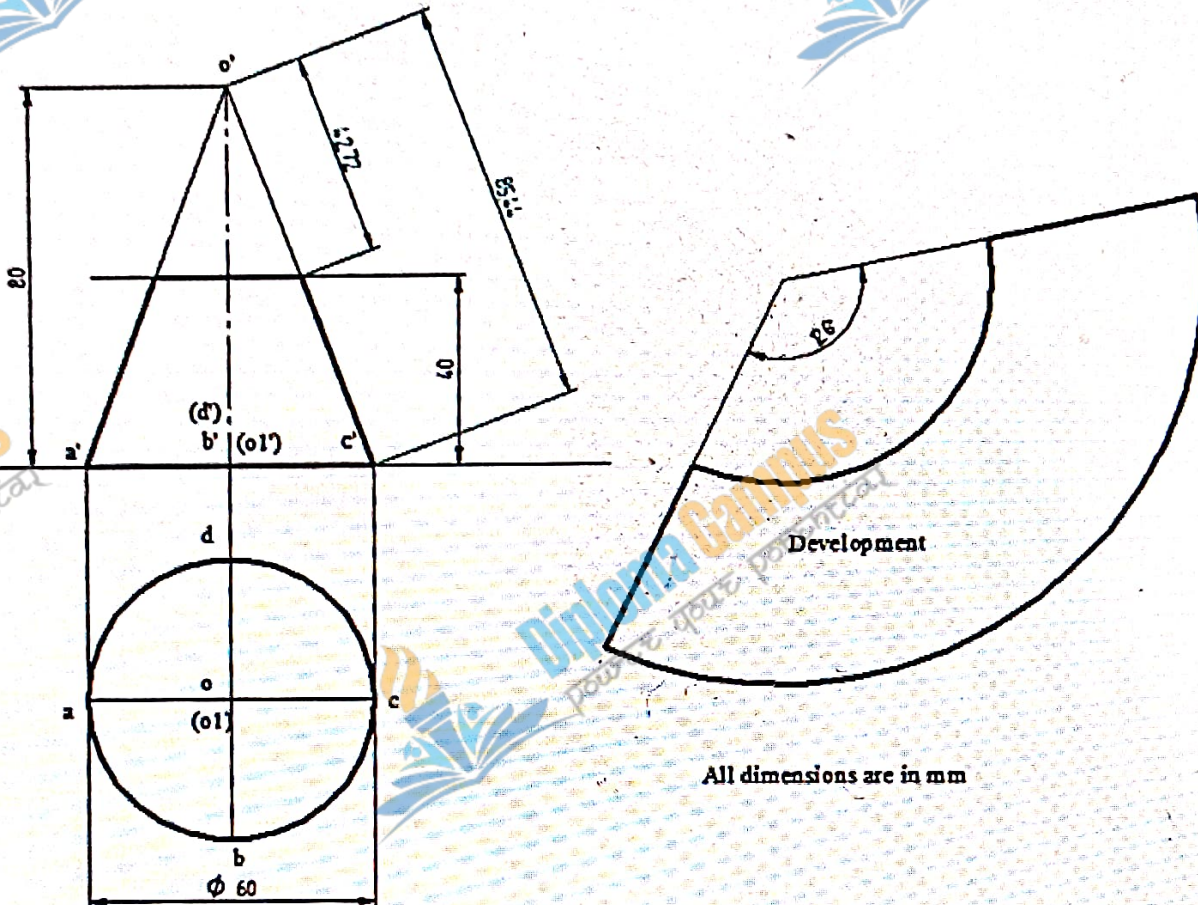


- Continue the same procedure of marking, bending, forming



Visual Watermark

Development of Frustum of Cone

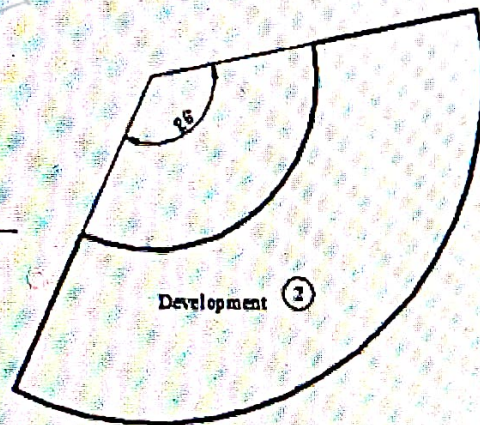
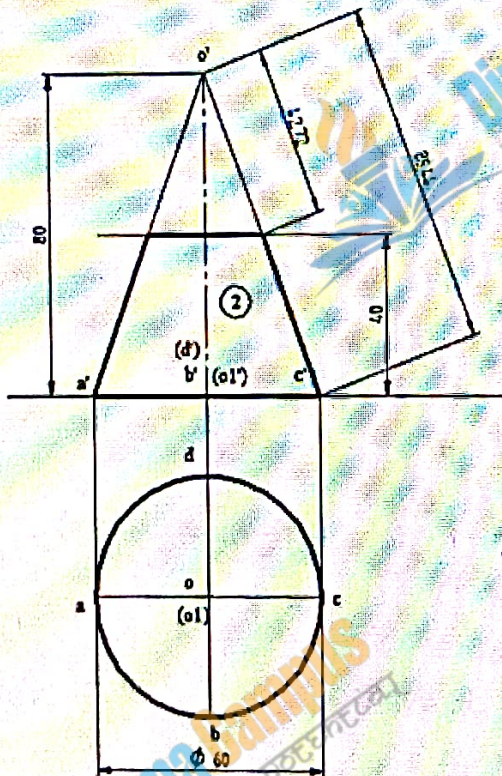
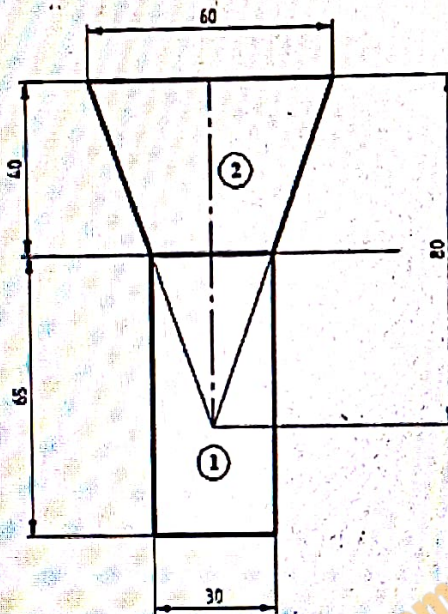


Procedure:

Draw the top view and front view of the right circular cone of base dia. 60mm. A section plane cuts perpendicular to the axis of the cone at 40mm height above the base.
With 'O' as center and radius equal to slant generator length (O'C' or O'A'), draw an arc.

Find $\theta = (r/R) \times 360^\circ$ where r = Radius of the base circle of the cone, R = Length of slant generator of the cone and 360° = Angle subtended to cut the arc.

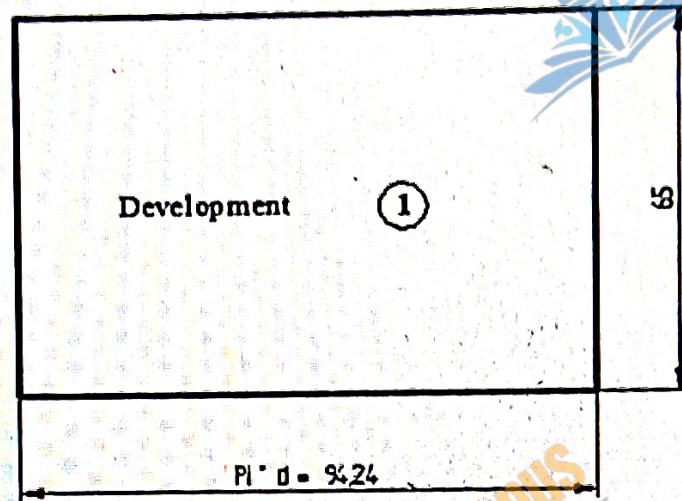
Development of Funnel



All dimensions are in mm



Visual Watermark



Procedure for preparing a funnel

Draw the top view and front view of the right circular cone of base dia. 60mm. A section plane cuts perpendicular to the axis of the cone at 40mm height above the base. With 'O' as center and radius equal to slant generator length (O'C' or O'A'), draw an arc.

Find $\theta = (r/R) \times 360^\circ$ where r = Radius of the base circle of the cone, R = Length of slant generator of the cone and 360° = Angle subtended to cut the arc.

- Draw the top view and front view of the cylinder of base dia. 30mm and bottom of funnel is beveled off completely to 40° .
- Develop the cylinder of base diameter 30 mm and height 65mm.
- Circumference $= 2\pi r$, where r = Radius of cylinder
- Cut the sheet along the line according to shape of the development.
- Fold 5 mm extra allowance in clockwise and anti-clockwise directions by keeping hacksaw blade thickness and pressed.
- Remove the hacksaw blade, Bend the main body using the stake and lock the end joint.
- Solder the two parts with electric soldering.

Manufacturing Process

EXPT NO: 02

ROUND PIPE

AIM: To make a cylindrical pipe as per required dimensions.

MATERIALS REQUIRED:

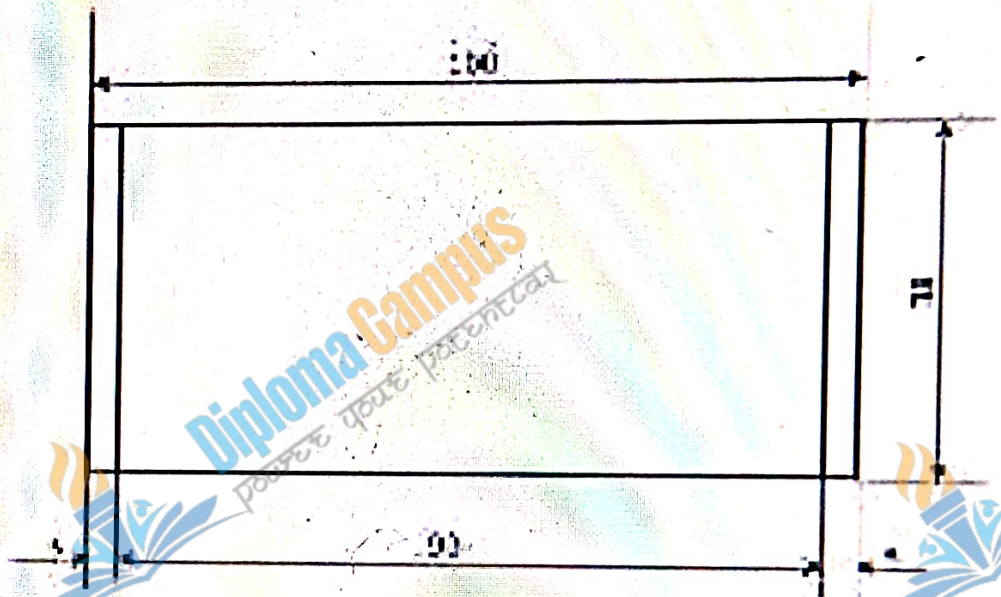
Galvanized Iron sheet (100*38*26 gauge)

TOOLS AND EQUIPMENT REQUIRED:

1. Steel rule
2. Flat file
3. Scriber
4. Try square
5. Snips
6. Dot punch
7. Stakes
8. Mallet

OPERATIONS TO BE CARRIED OUT:

1. Planning
2. Marking
3. Cutting
4. Bending
5. Seaming
6. Soldering



PROCEDURE:

1. The size of the given sheet is checked with steel rule.
2. Mark the measurement and make the development surface sketch diagram.
3. The layout of the cylindrical shape pipe is marked on the given sheet.
4. The sheet is bent to the required shape using stakes and mallet.
5. Now the edges are slightly bent to one is one side and the other is opposite side, using stakes and mallet.
6. Join both the ends with in a cylindrical shape.

PRECAUTIONS:

1. Care must be taken while cutting snips.
2. Care must be taken while bending and joining.

RESULT:

The cylindrical shape is made successfully.